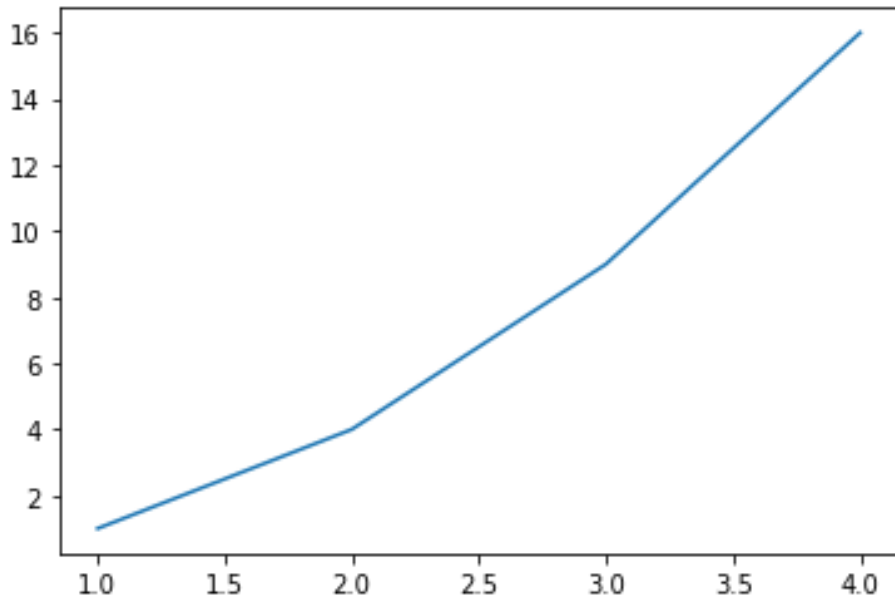


1.2 맷플롯립으로 그래프그리기

In []:

```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4]  
y = [1, 4, 9, 16]  
plt.plot(x,y)  
plt.show()
```

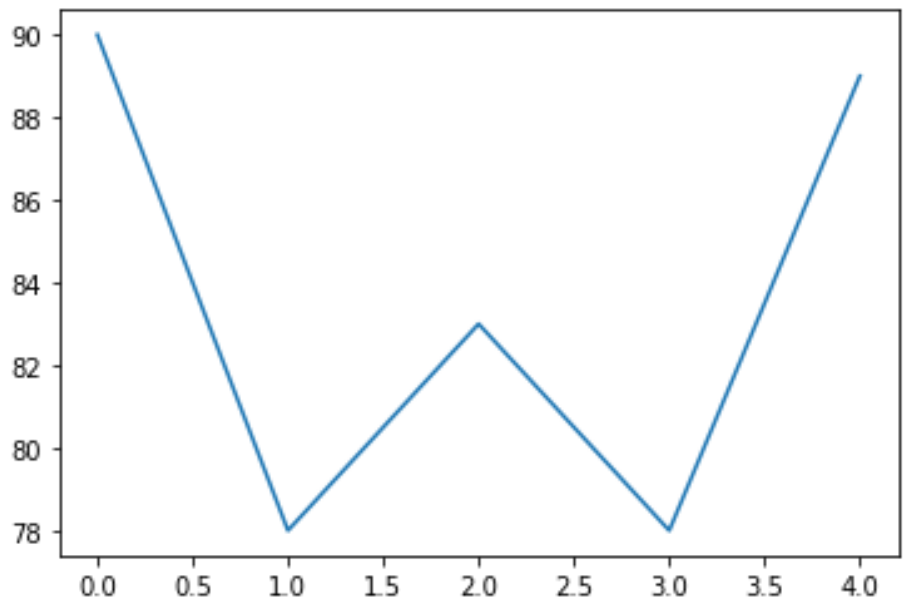


```
In [ ]:
```

```
import matplotlib.pyplot as plt  
import pandas as pd
```

```
df=pd.read_excel('data1.xlsx')
```

```
plt.plot(df['kor'])  
plt.show()
```

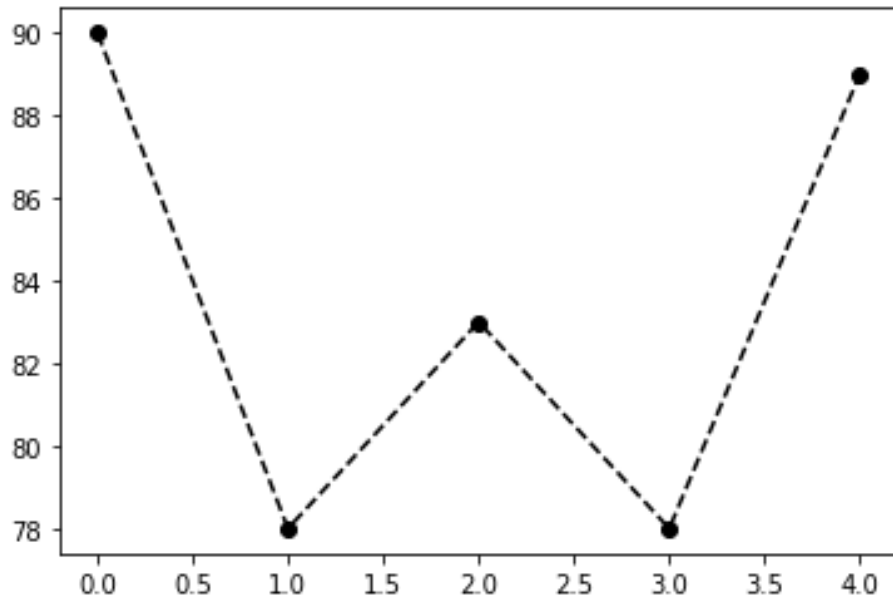


```
In [ ]:
```

```
import matplotlib.pyplot as plt
import pandas as pd

df=pd.read_excel('data1.xlsx')

plt.plot(df['kor'], 'k--',marker='o')
plt.show()
```

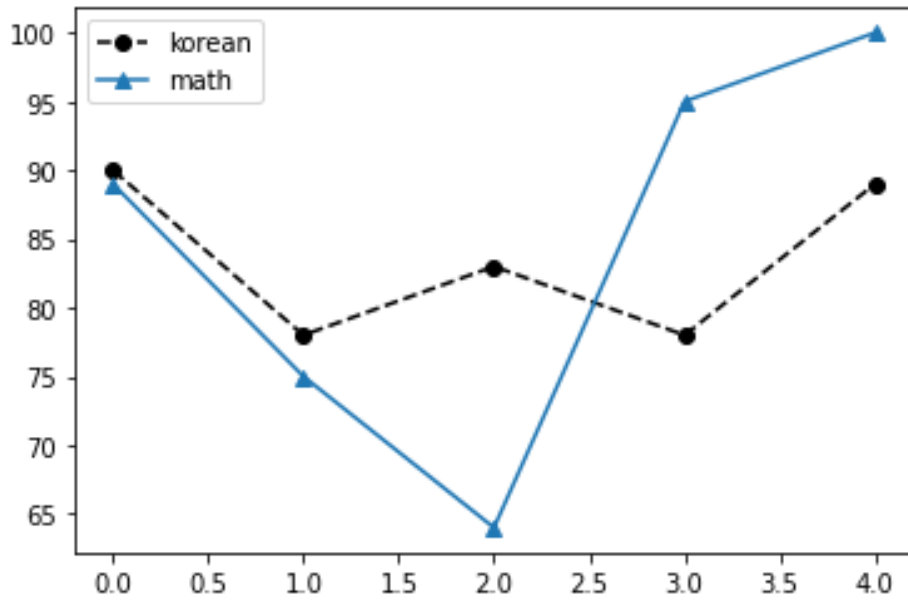


```
In [ ]:
```

```
import matplotlib.pyplot as plt  
import pandas as pd
```

```
df=pd.read_excel('data1.xlsx')
```

```
plt.plot(df['kor'], 'k--', marker='o', label='korean')  
plt.plot(df['math'], marker='^',label='math')  
plt.legend()  
plt.show()
```

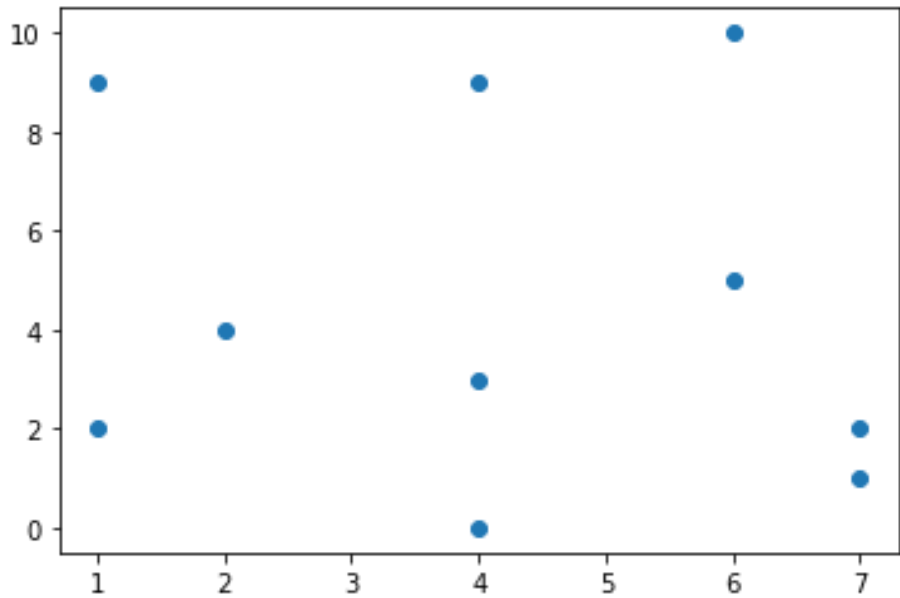


In []:

```
import matplotlib.pyplot as plt
```

```
x = [6,2,1,1,4,4,6,4,7,7]  
y = [5,4,2,9,9,0,10,3,1,2]
```

```
plt.scatter(x,y)  
plt.show()
```

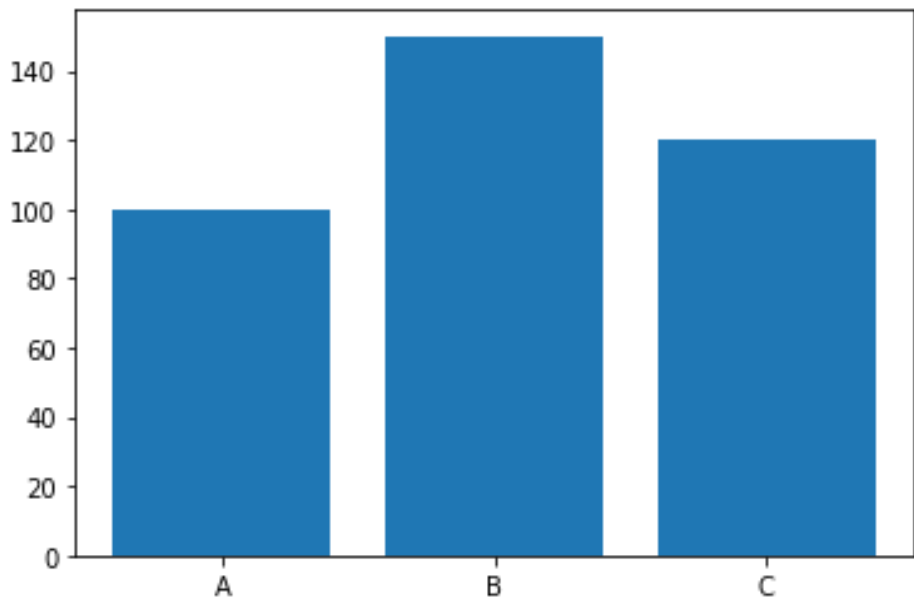


In []:

```
import matplotlib.pyplot as plt
```

```
x = ['A', 'B', 'C']  
y = [100, 150, 120]
```

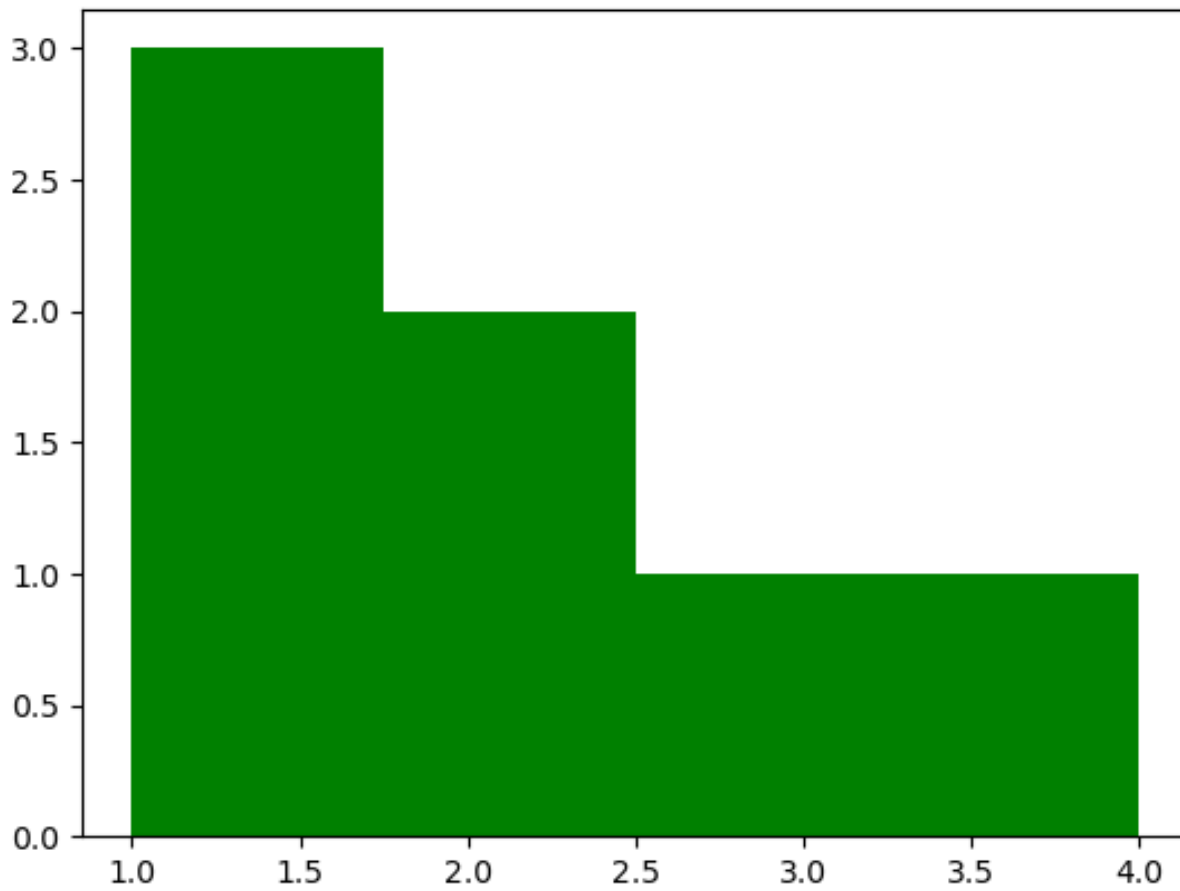
```
plt.bar(x,y)  
plt.show()
```



```
In [1]:
```

```
import matplotlib.pyplot as plt

x= [1,1,1,2,3,2,4]
plt.hist(x, bins=4, color='green', align='mid')
plt.show()
```

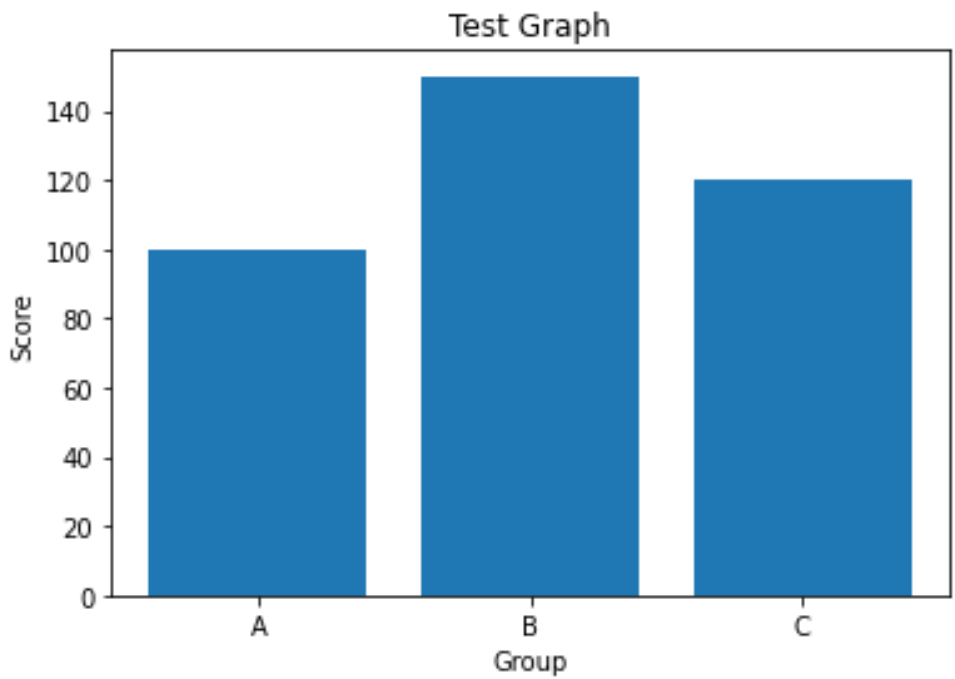


```
In [ ]:
```

```
import matplotlib.pyplot as plt
```

```
x = ['A', 'B', 'C']  
y = [100, 150, 120]
```

```
plt.bar(x,y)  
plt.title('Test Graph')  
plt.xlabel('Group')  
plt.ylabel('Score')  
plt.show()
```



```
In [ ]:
```

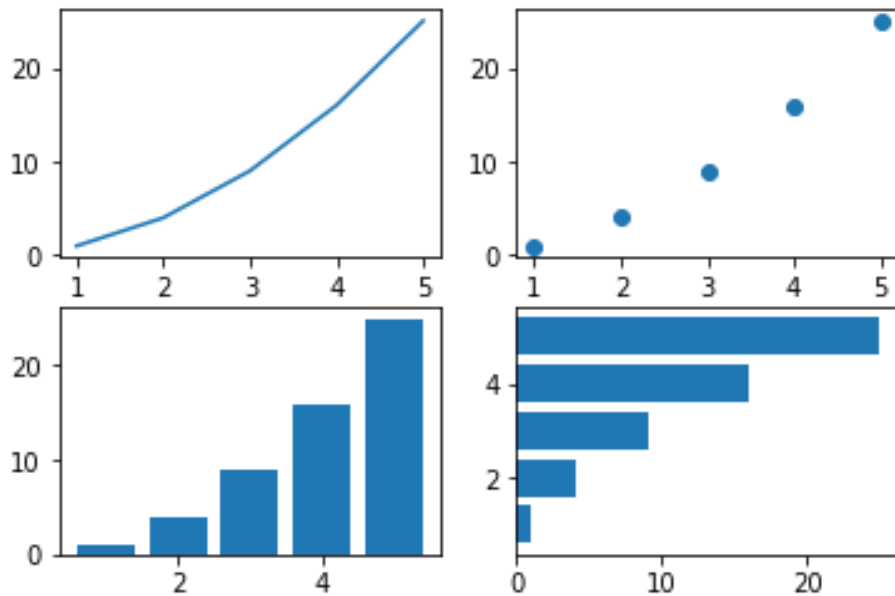
```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4, 5]  
y = [1, 4, 9, 16, 25]
```

```
fig, axs = plt.subplots(2,2)
```

```
axs[0,0].plot(x,y)  
axs[0,1].scatter(x,y)  
axs[1,0].bar(x,y)  
axs[1,1].barh(x,y)  
fig.suptitle('Drawing Graph')  
plt.show()
```

Drawing Graph



In []:

```
!sudo apt-get install -y fonts-nanum  
!sudo fc-cache -fv  
!rm ~/.cache/matplotlib -rf
```

Reading package lists... Done
Building dependency tree
Reading state information... Done
The following package was automatically installed and is no longer required:
libnvidia-common-460
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
fonts-nanum
0 upgraded, 1 newly installed, 0 to remove and 27 not upgraded.
Need to get 9,604 kB of archives.
After this operation, 29.5 MB of additional disk space will be used.
Get:1 <http://archive.ubuntu.com/ubuntu> (<http://archive.ubuntu.com/ubuntu>) bionic/universe amd64 fonts-nanum all 20170925-1 [9,604 kB]
Fetched 9,604 kB in 3s (3,452 kB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 76, <> line 1.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (This frontend requires a controlling tty.)
debconf: falling back to frontend: Teletype
dpkg-preconfigure: unable to re-open stdin:
Selecting previously unselected package fonts-nanum.
(Reading database ... 123942 files and directories currently installed.)
Preparing to unpack .../fonts-nanum_20170925-1_all.deb ...
Unpacking fonts-nanum (20170925-1) ...
Setting up fonts-nanum (20170925-1) ...
Processing triggers for fontconfig (2.12.6-0ubuntu2) ...
/usr/share/fonts: caching, new cache contents: 0 fonts, 1 dirs
/usr/share/fonts/truetype: caching, new cache contents: 0 fonts, 3 dirs
/usr/share/fonts/truetype/humor-sans: caching, new cache contents: 1 fonts, 0 dirs
/usr/share/fonts/truetype/liberation: caching, new cache contents: 16 fonts, 0 dirs
/usr/share/fonts/truetype/nanum: caching, new cache contents: 10 fonts, 0 dirs
/usr/local/share/fonts: caching, new cache contents: 0 fonts, 0 dirs
/root/.local/share/fonts: skipping, no such directory
/root/.fonts: skipping, no such directory
/var/cache/fontconfig: cleaning cache directory
/root/.cache/fontconfig: not cleaning non-existent cache directory
/root/.fontconfig: not cleaning non-existent cache directory
fc-cache: succeeded

```
In [ ]:
```

```
import matplotlib.pyplot as plt
```

```
plt.rc('font', family='NanumGothic')
```

```
x = [1, 2, 3, 4]
```

```
y = [10, 4, 15, 9]
```

```
plt.plot(x,y,label='국어')
```

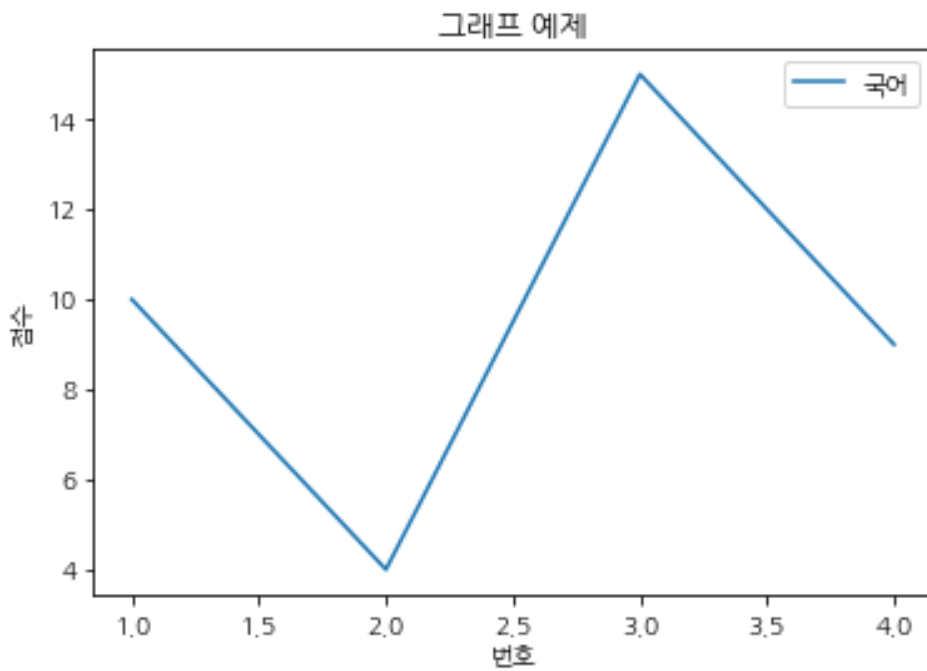
```
plt.title('그래프 예제')
```

```
plt.xlabel('번호')
```

```
plt.ylabel('점수')
```

```
plt.legend()
```

```
plt.show()
```



In [2]:

```
# 주피터노트북 전용 코드 : 컴퓨터에 설치된 폰트 확인
import matplotlib.font_manager as fm

fonts = sorted([a.name for a in fm.fontManager.ttflist])
fonts
```

Out[2]:

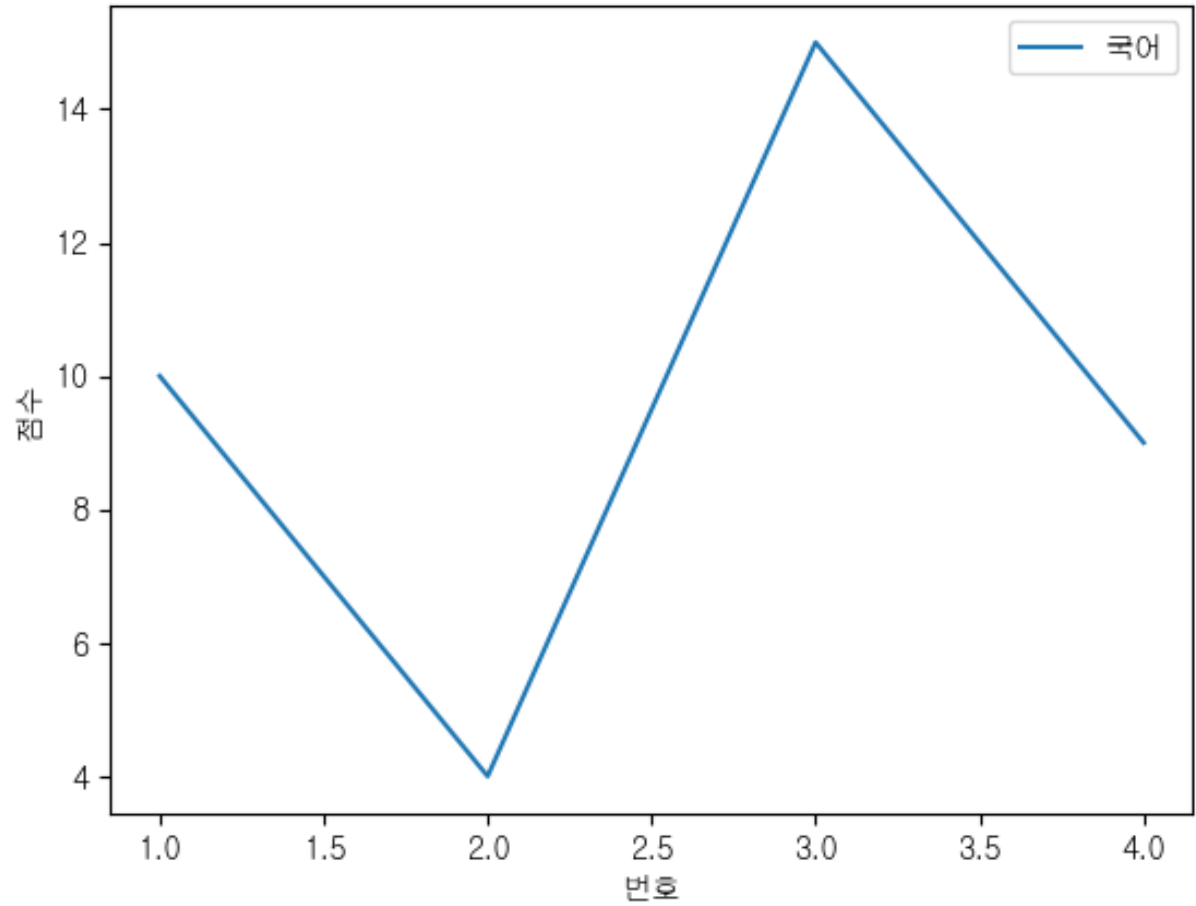
[illegible]

In [3]:

```
# 주피터노트북 전용 코드 : 컴퓨터에 설치된 폰트인 굴림체(Gulim)로 폰트 지정
import matplotlib.pyplot as plt

plt.rc('font', family='Gulim')
x = [1, 2, 3, 4]
y = [10, 4, 15, 9]
plt.plot(x,y,label='국어')
plt.title('그래프 예제')
plt.xlabel('번호')
plt.ylabel('점수')
plt.legend()
plt.show()
```

그래프 예제



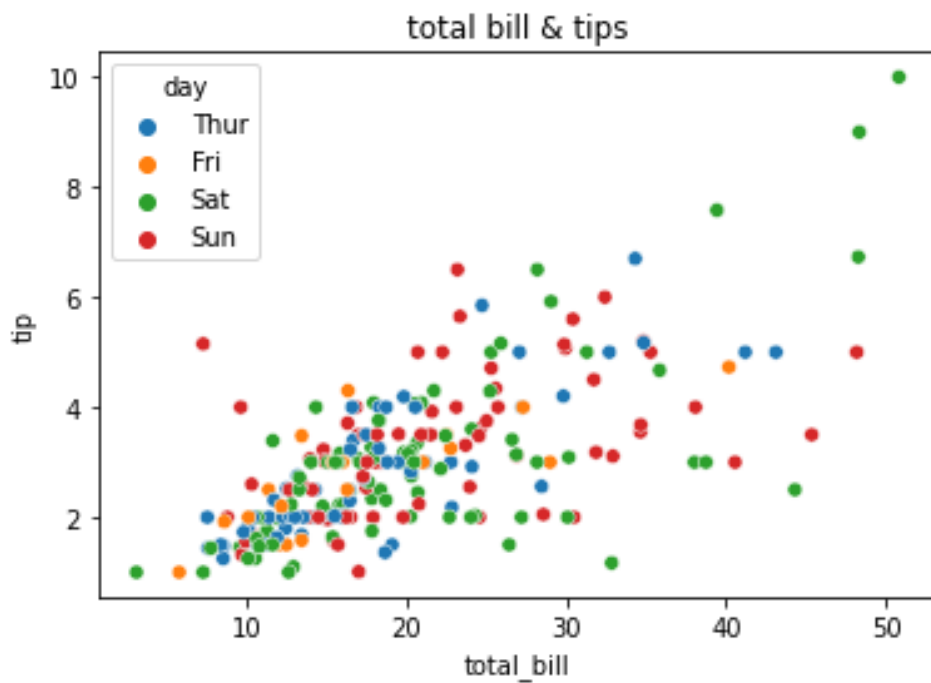
2.2시본으로 그래프 그리기

In []:

```
import matplotlib.pyplot as plt
import seaborn as sns

tips_data = sns.load_dataset('tips')

sns.scatterplot(x='total_bill', y='tip',
                hue='day', data=tips_data)
plt.title('total bill & tips')
plt.show()
```

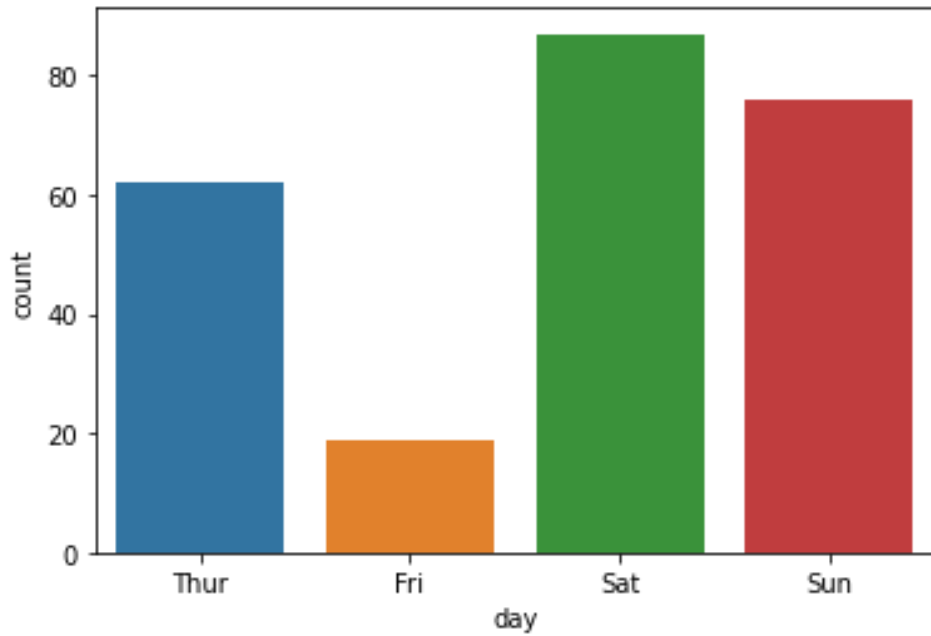


```
In [ ]:
```

```
import matplotlib.pyplot as plt
import seaborn as sns

tips_data = sns.load_dataset('tips')

sns.countplot(x='day', data=tips_data)
plt.show()
```

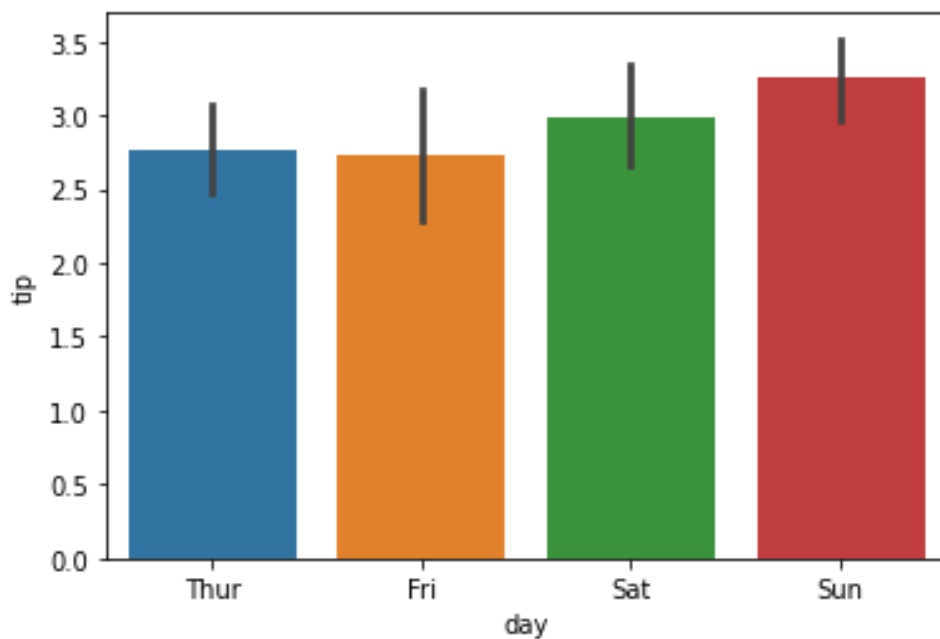


```
In [ ]:
```

```
import matplotlib.pyplot as plt
import seaborn as sns

tips_data = sns.load_dataset('tips')

sns.barplot(x='day', y='tip', data=tips_data)
plt.show()
```



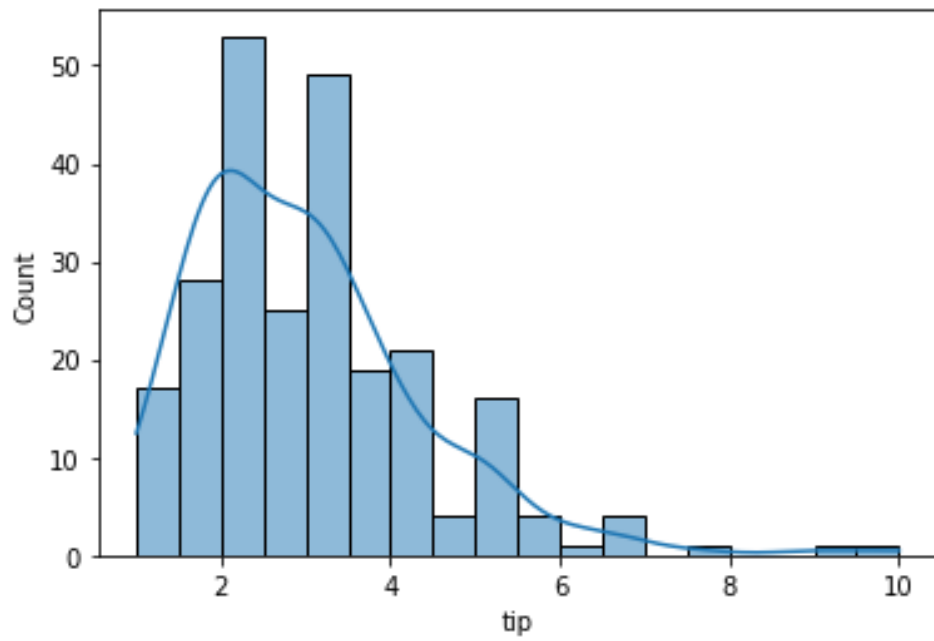
```
In [ ]:
```

```
import matplotlib.pyplot as plt
import seaborn as sns

tips_data = sns.load_dataset('tips')

sns.histplot(x='tip', data=tips_data,
             kde=True)

plt.show()
```

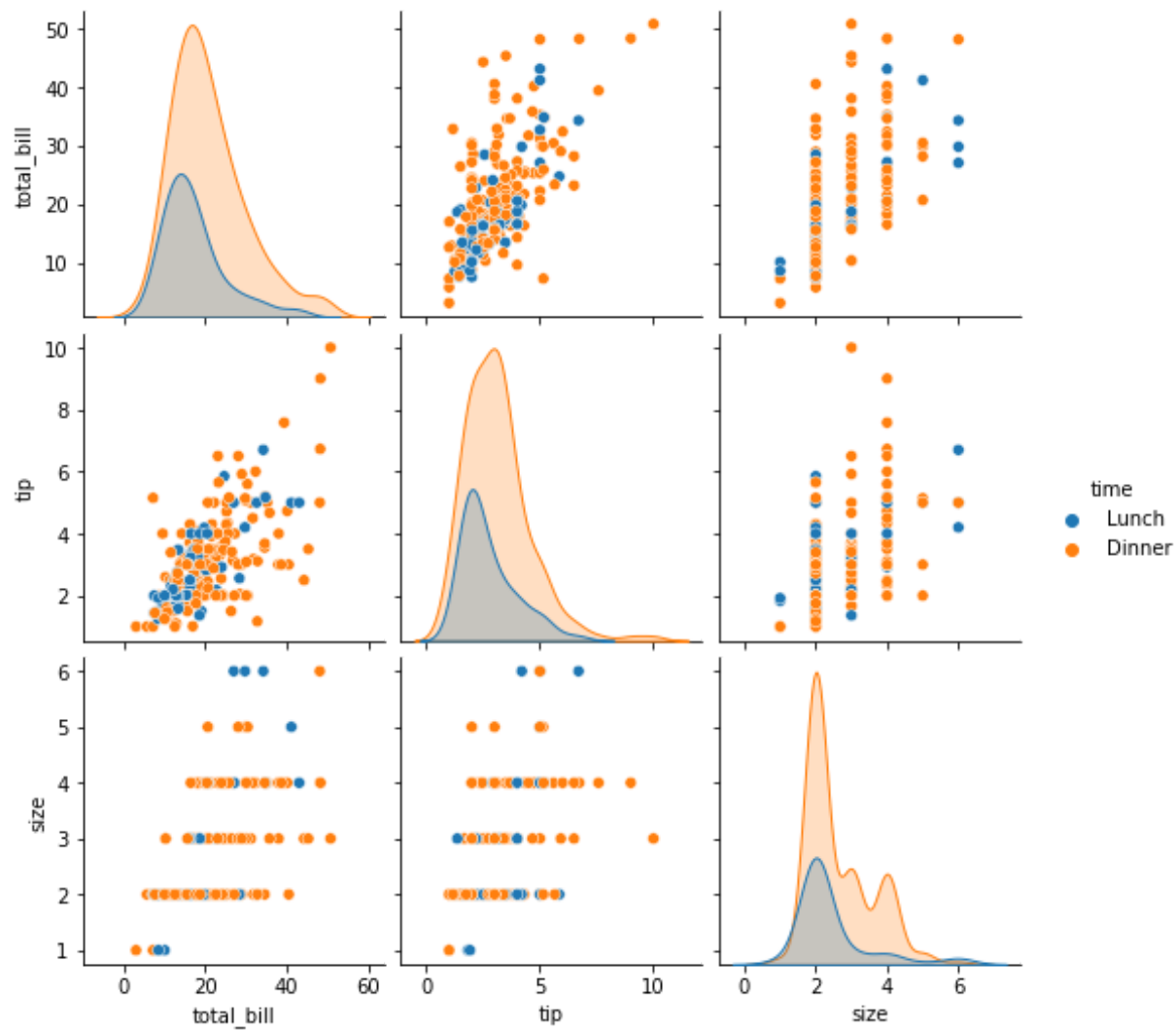


```
In [ ]:
```

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
tips_data = sns.load_dataset('tips')
```

```
sns.pairplot(tips_data,hue='time')
plt.show()
```

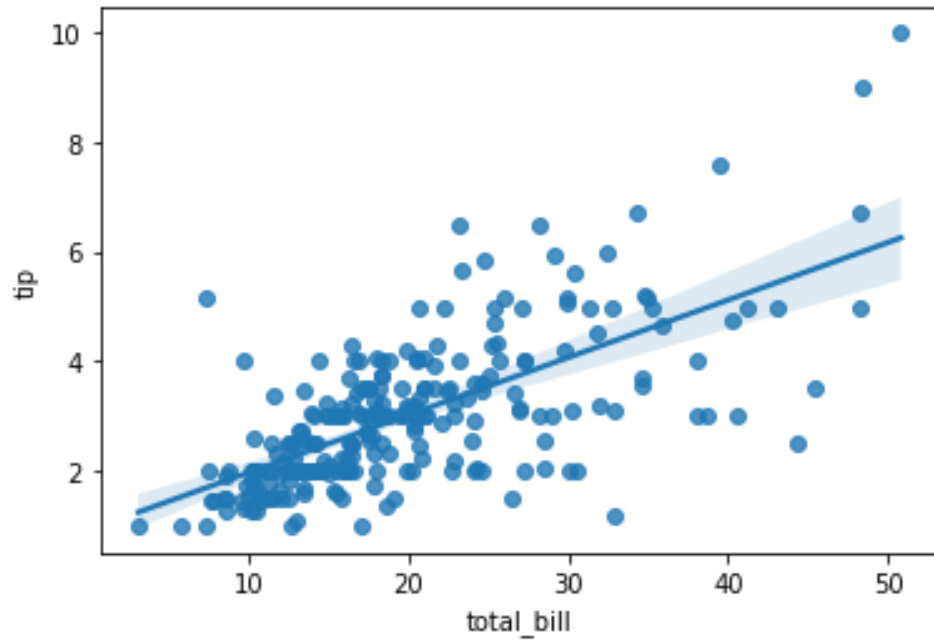


In []:

```
import matplotlib.pyplot as plt
import seaborn as sns

tips_data = sns.load_dataset('tips')

sns.regplot(x='total_bill',y='tip',
            data=tips_data)
plt.show()
```

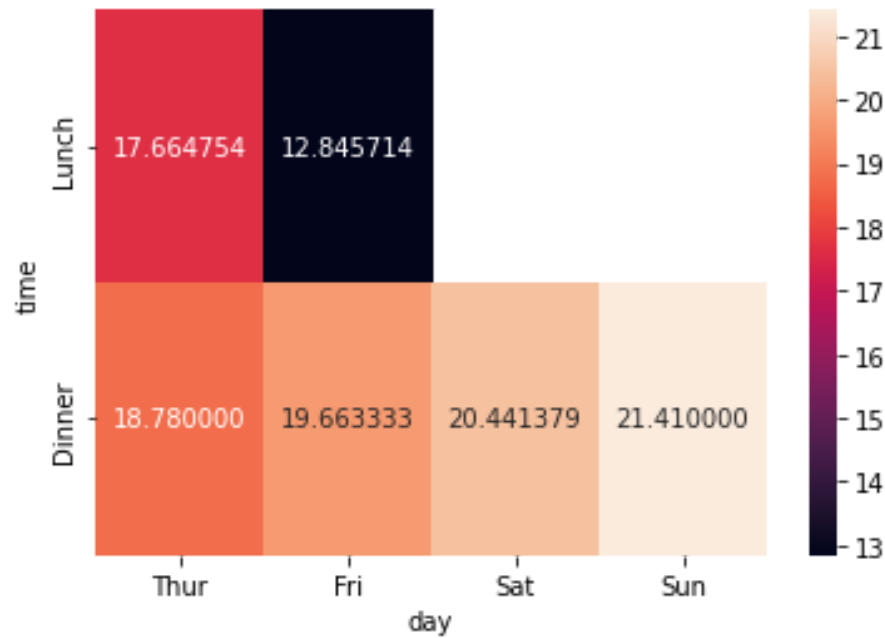


In []:

```
tips_data
```

In []:

```
bill_pivot = tips_data.pivot_table(index='time',
                                    columns='day',values='total_bill')
sns.heatmap(bill_pivot, annot=True, fmt='f')
plt.show()
```



In []:

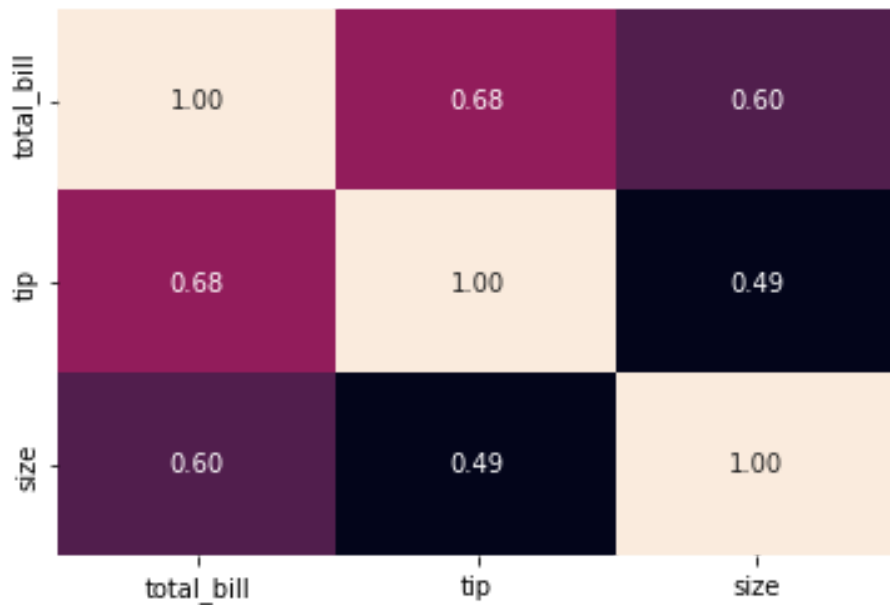
```
tips_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   total_bill  244 non-null   float64
1   tip         244 non-null   float64
2   sex        244 non-null   category
3   smoker     244 non-null   category
4   day        244 non-null   category
5   time       244 non-null   category
6   size       244 non-null   int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
```

In []:

```
import matplotlib.pyplot as plt
import seaborn as sns

tips_data = sns.load_dataset('tips')
df= tips_data.corr()
sns.heatmap(data=df,annot=True,
            fmt='.2f', cbar=False)
plt.show()
```



In []: